



Convolutional Neural Networks

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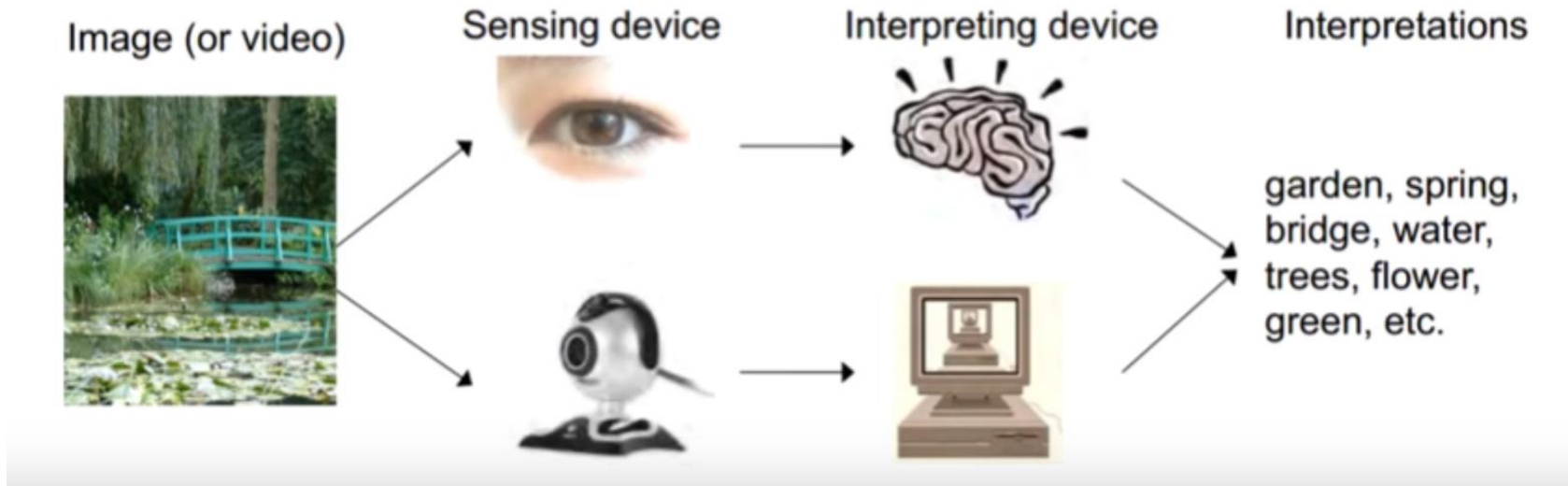
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Computer Vision



Computer vision is a field of artificial intelligence (AI) that enables computers and systems to interpret, analyze, and derive meaningful information from digital images, videos, and other visual inputs.

Computer Vision – Problem Types

Classification



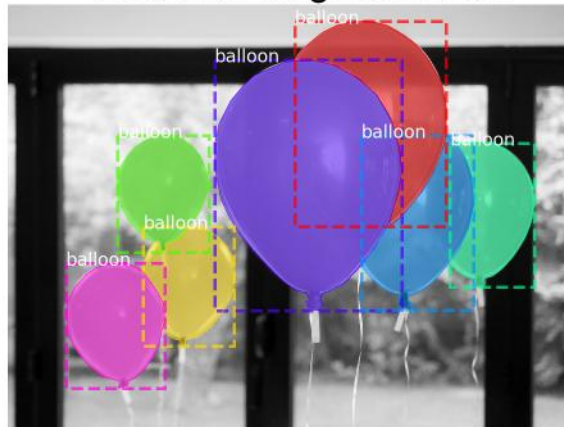
Semantic Segmentation



Object Detection



Instance Segmentation



Classification: There is a balloon in this image

Object detection: There are 7 balloons in this image at these locations. We're starting to account for objects that overlap

Semantic segmentation: These are all the balloon pixels

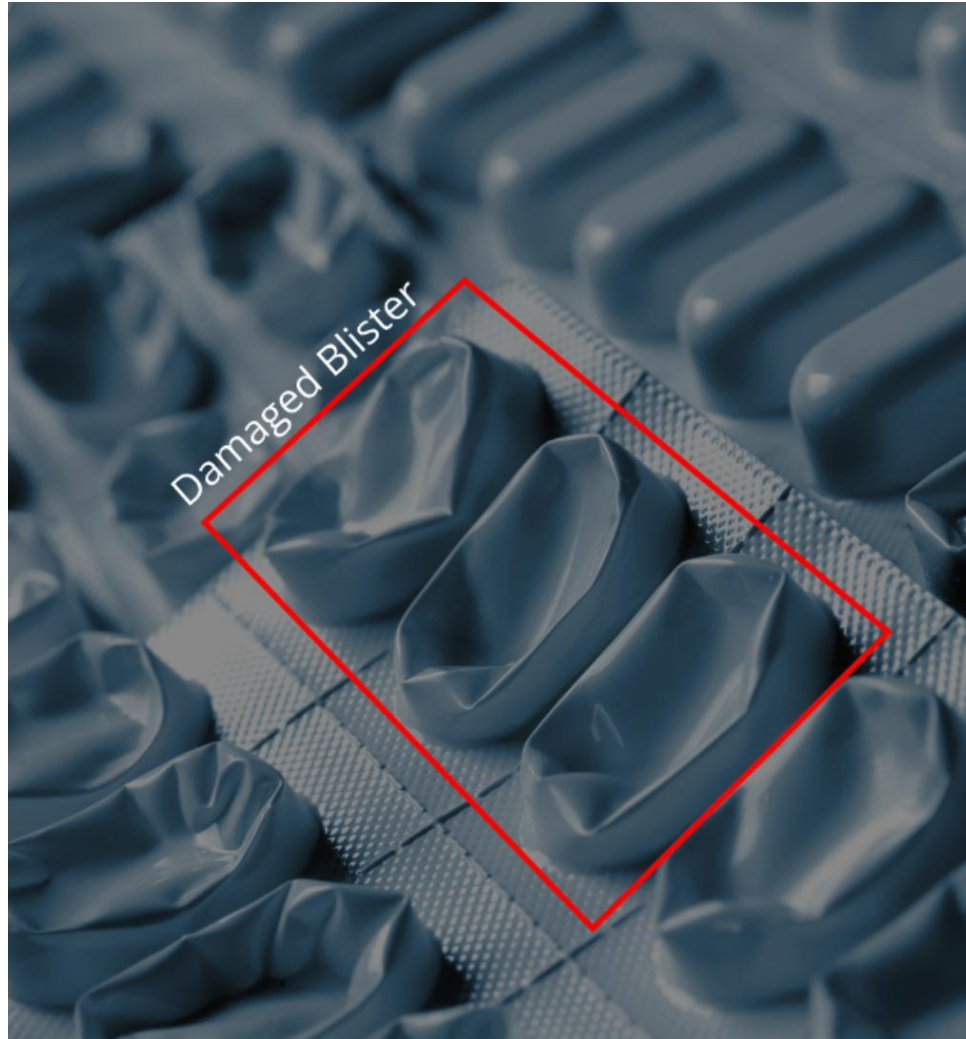
Instance segmentation: There are 7 balloons at these locations, and these are the pixels that belong to each one

Visual Search



Source: <https://www.leewayhertz.com/ai-in-visual-search/>

Defect Detection



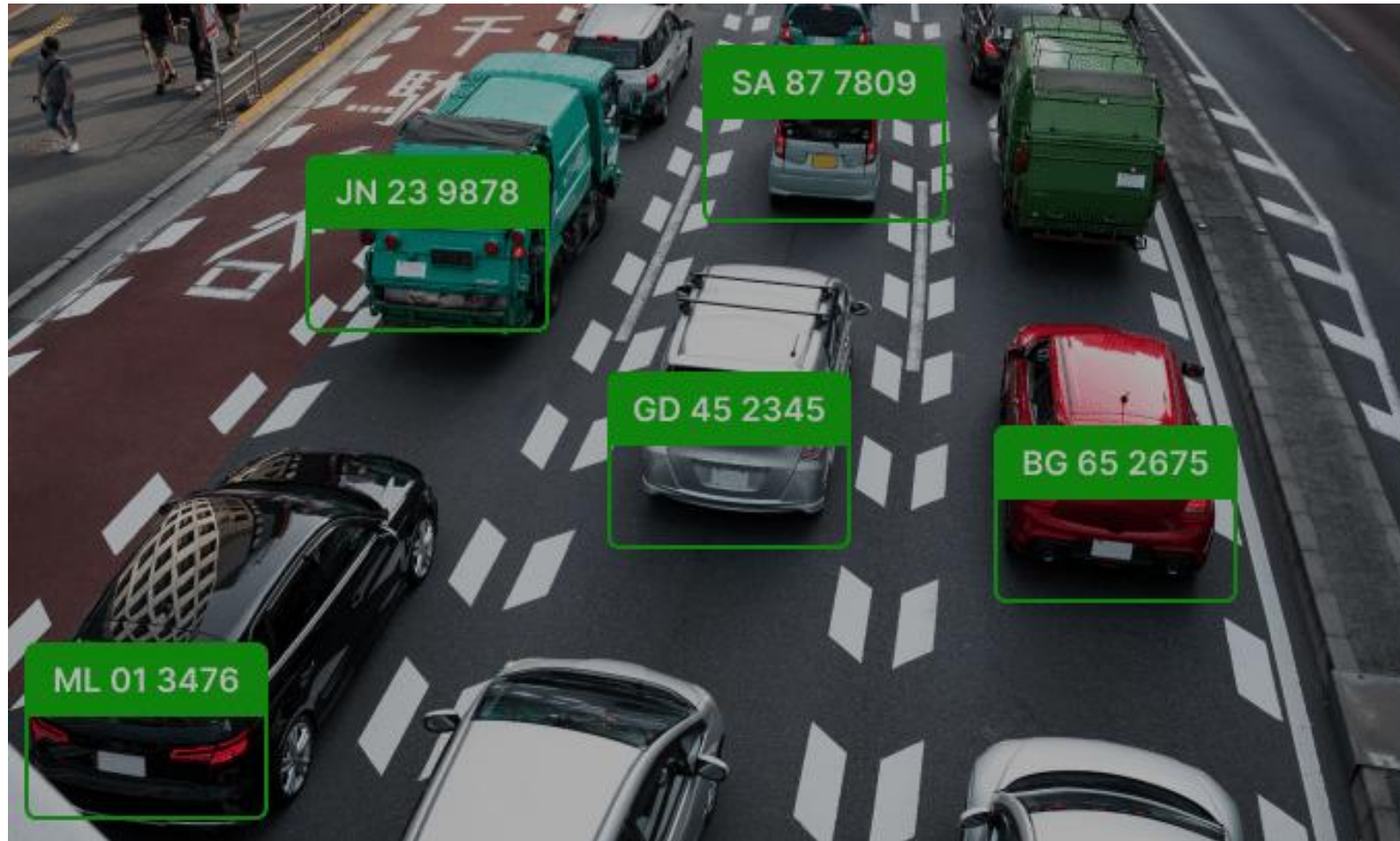
Source: <https://imagevision.ai/applications/defect-detection/>

Crop Disease Detection



Source: <https://blog.google/technology/ai/ai-takes-root-helping-farmers-identify-diseased-plants/>

License Plate Recognition



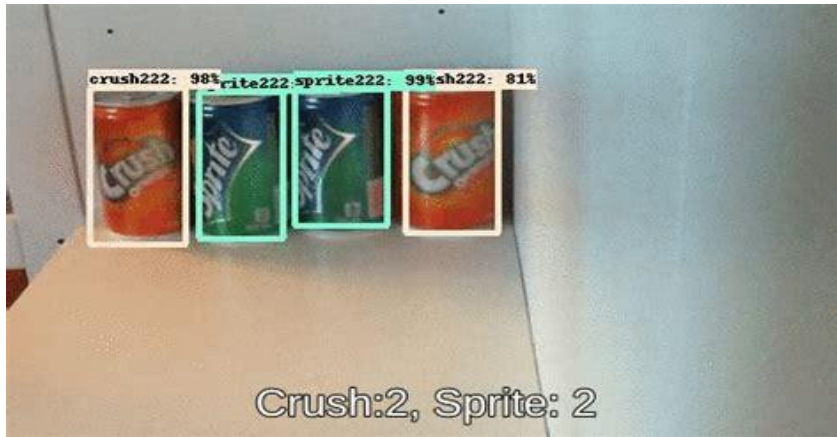
Source: <https://nextbrain.ca/exploring-automatic-number-plate-recognition-system-harnessing-computer-vision/>

Facial Recognition (Example: DigiYatra)



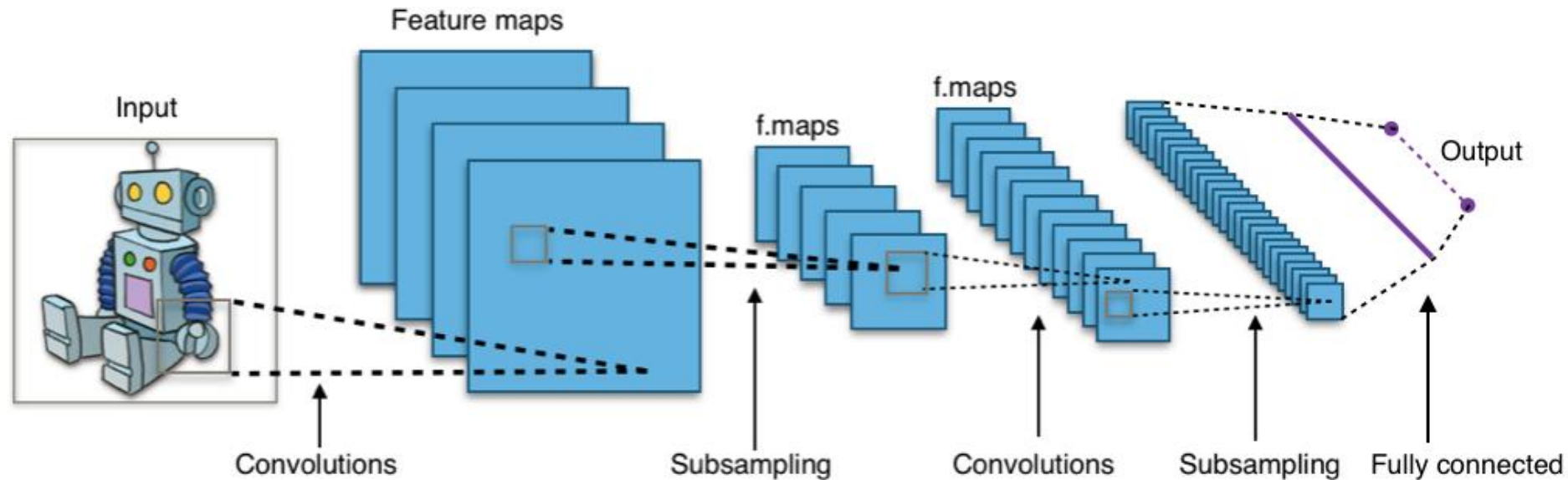
Source: <https://content.techgig.com/technology/facial-recognition-technology-set-to-debut-at-chennai-airport/articleshow/110444885.cms>

Object Detection Applications



Convolutional Neural Network

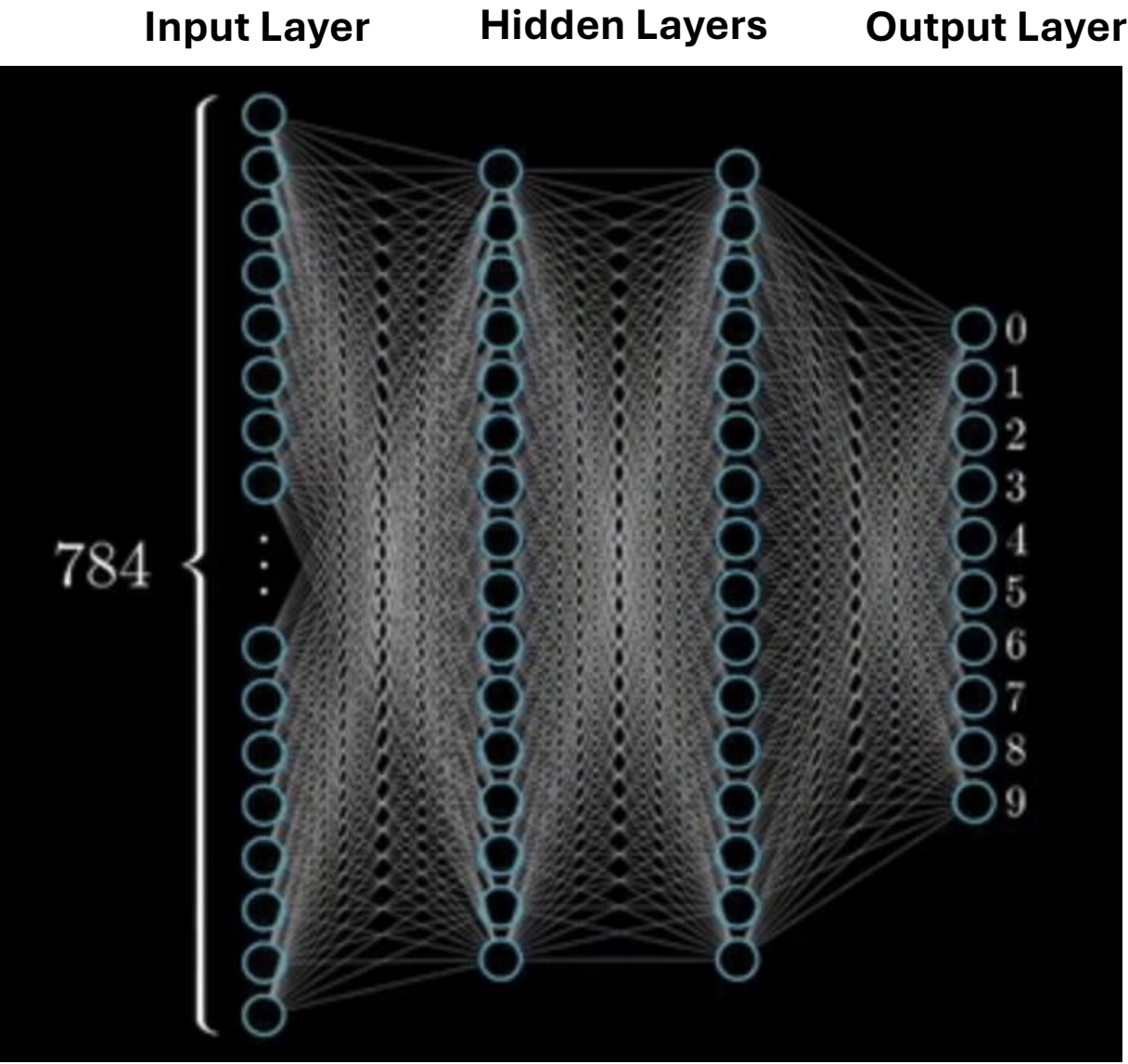
- CNN - Convolutional Neural Network
 - Feed-forward artificial neural network
 - Convolutional networks were inspired by biological processes



Entering The Pixel Level Data To An Artificial Neural Network

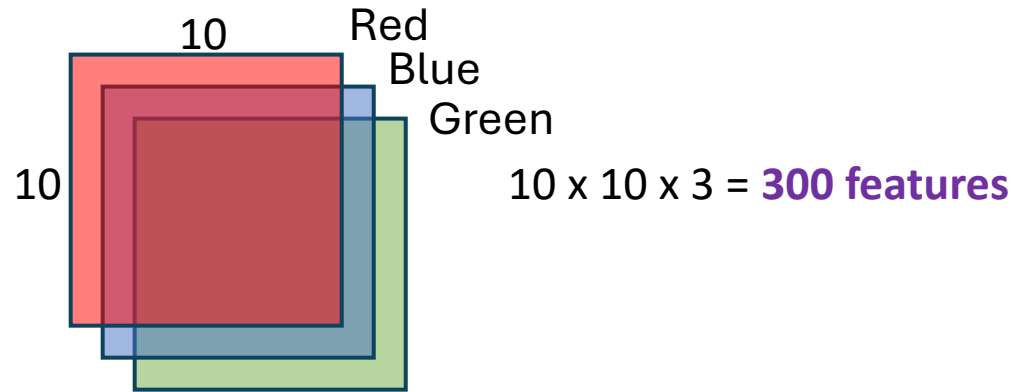
Pixel Level Data

[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0]
[	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	13	73	0	0	1	4	0	0	0	0	1	1	1	0]
[	0	0	0	0	0	0	0	0	0	0	0	0	3	0	36	136	127	62	54	0	0	0	1	3	4	0	0	3]	
[	0	0	0	0	0	0	0	0	0	0	0	0	6	0	102	204	176	134	144	123	23	0	0	0	0	12	10	0]	
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	155	236	207	178	107	156	161	109	64	23	77	130	72	15]	
[	0	0	0	0	0	0	0	0	0	0	0	1	0	69	207	223	218	216	216	163	127	121	122	146	141	88	172	66]	
[	0	0	0	0	0	0	0	0	0	0	1	1	1	0	200	232	232	233	229	223	223	215	213	164	127	123	196	229	0]
[	0	0	0	0	0	0	0	0	0	0	0	0	0	183	225	216	223	228	235	227	224	222	224	221	223	245	173	0]	
[	0	0	0	0	0	0	0	0	0	0	0	0	0	193	228	218	213	198	180	212	210	211	213	223	220	243	202	0]	
[	0	0	0	0	0	0	0	0	0	1	3	0	12	219	220	212	218	192	169	227	208	218	224	212	226	197	209	52]	
[	0	0	0	0	0	0	0	0	0	0	6	0	99	244	222	220	218	203	198	221	215	213	222	220	245	119	167	56]	
[	0	0	0	0	0	0	0	0	0	4	0	0	55	236	228	230	228	240	232	213	218	223	234	217	217	209	92	0]	
[	0	0	1	4	6	7	2	0	0	0	0	0	237	226	217	223	222	219	222	221	216	223	229	215	218	255	77	0]	
[	0	3	0	0	0	0	0	0	0	62	145	204	228	207	213	221	218	208	211	218	224	223	219	215	224	244	159	0]	
[	0	0	0	0	18	44	82	107	189	228	220	222	217	226	200	205	211	230	224	234	176	188	250	248	233	238	215	0]	
[	0	57	187	208	224	221	224	208	204	214	208	209	200	159	245	193	206	223	255	255	221	234	221	211	220	232	246	0]	
[	3	202	228	224	221	211	211	214	205	205	205	220	240	80	150	255	229	221	188	154	191	210	204	209	222	228	225	0]	
[	98	233	198	210	222	229	229	234	249	220	194	215	217	241	65	73	106	117	168	219	221	215	217	223	223	224	229	29]	
[	75	204	212	204	193	205	211	225	216	185	197	206	198	213	240	195	227	245	239	223	218	212	209	222	220	221	230	67]	
[	48	203	183	194	213	197	185	190	194	192	202	214	219	221	220	236	225	216	199	206	186	181	177	172	181	205	206	115]	
[	0	122	219	193	179	171	183	196	204	210	213	207	211	210	200	196	194	191	195	191	198	192	176	156	167	177	210	92]	
[	0	0	74	189	212	191	175	172	175	181	185	188	189	188	193	198	204	209	210	210	211	188	188	194	192	216	170	0]	
[	2	0	0	0	66	200	222	237	239	242	246	243	244	221	220	193	191	179	182	182	181	176	166	168	99	58	0	0]	
[	0	0	0	0	0	0	0	40	61	44	72	41	35	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0]	
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0]	
[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0]	



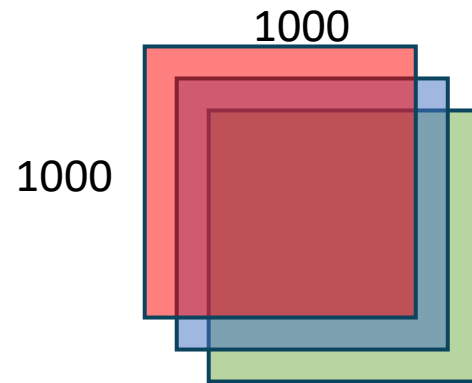
Color Image in Computer Vision

- For a 10 x 10 pixel color image



High Resolution Color Images

- For a 1000 x 1000 pixel color image



$$1000 \times 1000 \times 3 = \mathbf{3,000,000 \text{ features}}$$

High Resolution Color Images (Current Standards)



HD Image = 1280 x 720 pixels

FHD Image = 1920 x 1080 pixels

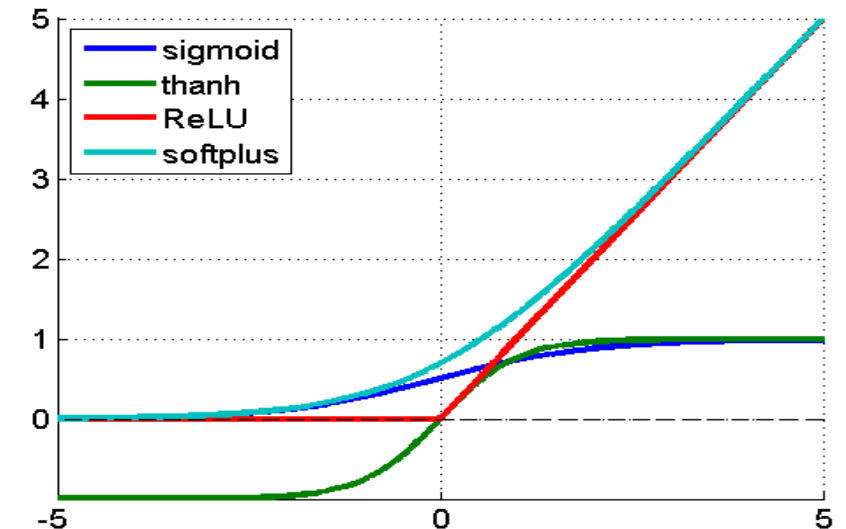
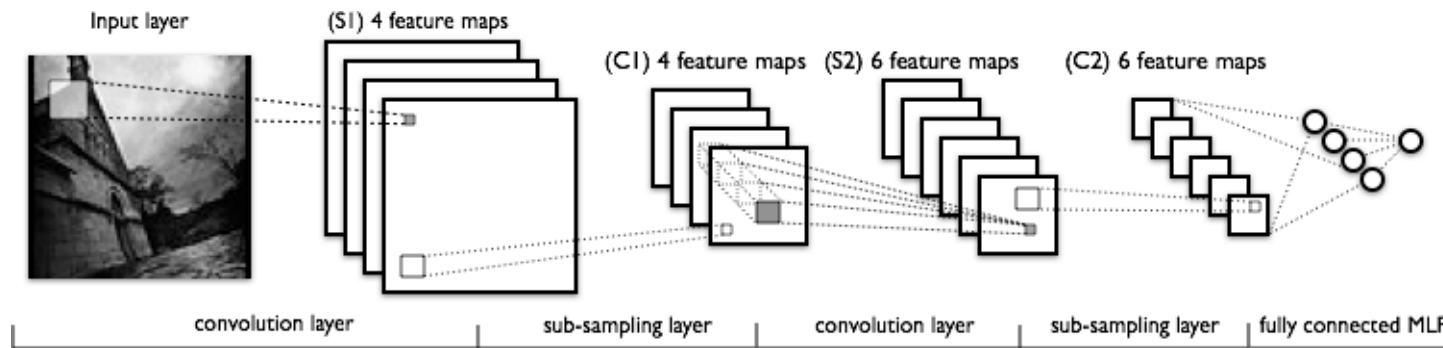
4K Image = 3840 x 2160 pixels

8K Image = 7680 x 4320 pixels

Why Convolution?

- Researchers spent 40 years on giving better inputs with filters
- Later focus is shifted to building architectures
- A convolution is an integral that expresses the amount of **overlap of one function** “g” as it is shifted over another function “f”. It therefore "blends" one function with another.
- Compared to fully connected network:
 - To reduce the number of operations and sparse connections
 - Each output value in a layer depends on sub set of inputs
 - Total number of parameters by parameter sharing with filters convolved over the image
 - Both for low level edge detector or high level feature detectors

Convolution operation



- Non-linearity is needed to learn complex (non-linear) representations of data, otherwise the NN would be just a linear function
- Most deep networks use ReLU - $\max(0, x)$, since it trains much faster, is more expressive than logistic function and prevents the gradient vanishing problem.

Convolution operation

Raw Image Pixel

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Filter or Kernel or Feature detector

1	0	1
0	1	0
1	0	1

1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

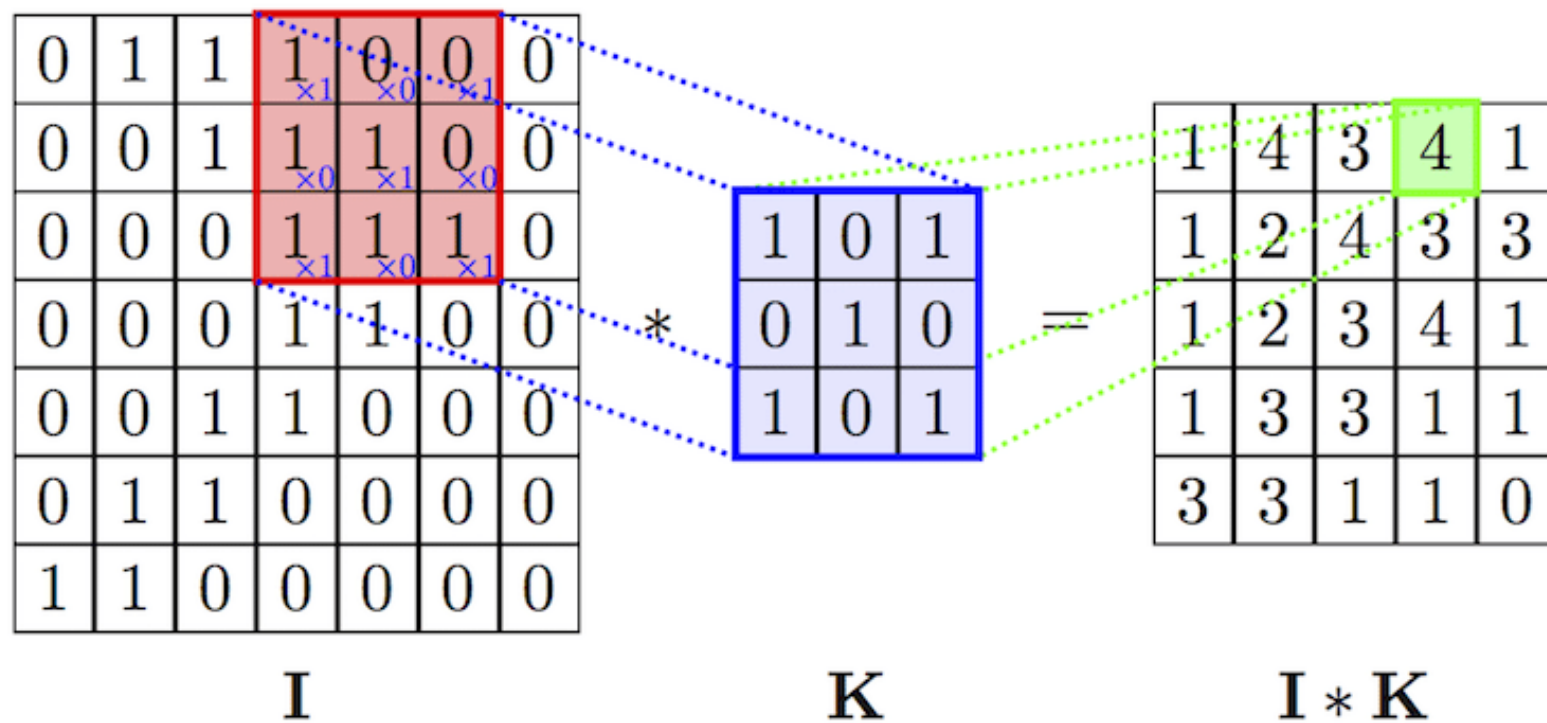
Image

4		

Convolved
Feature

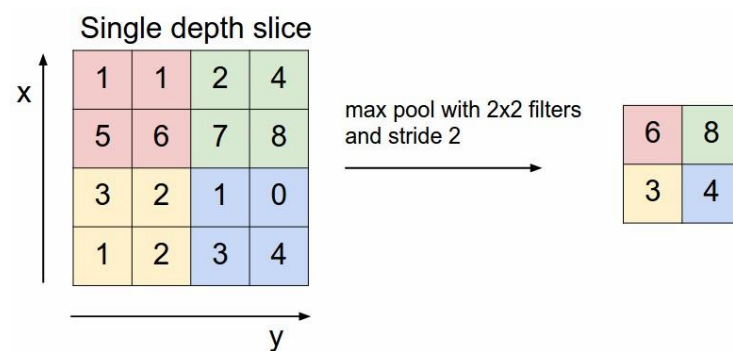
Convolved Feature or
Activation Map or the
Feature Map.

Convolution - 2D



MaxPooling

- Max pooling operation for temporal data.
- The max-pooling layer would **reduce the input Matrix into a down sampled size with max value for each block**. Please refer to the image shown below for an example.



Convolutional Neural Network

An image input constitutes a 3-dimensional structure called the *Input Volume* (255x255x3).

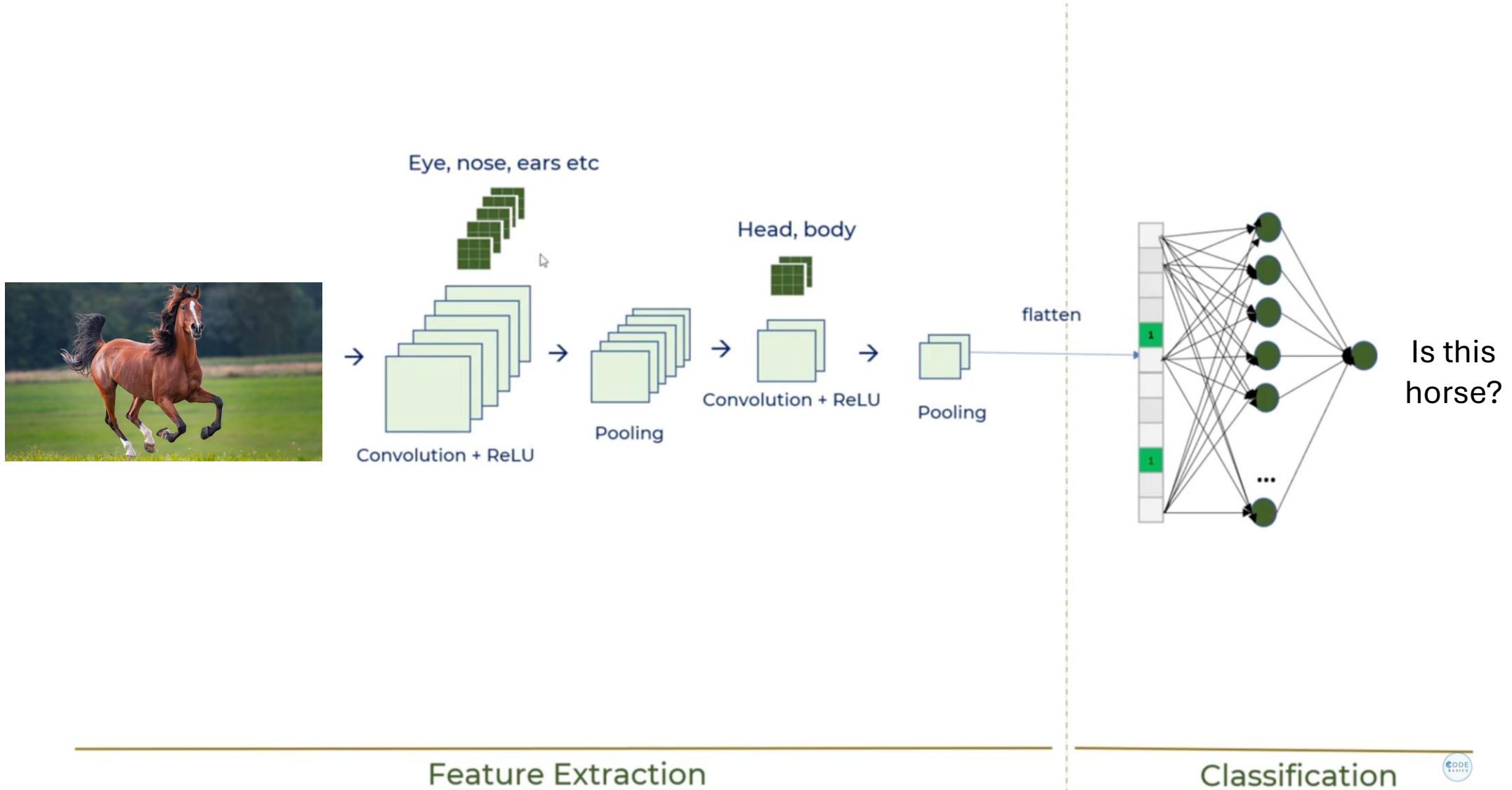
CNN's use **filters as kernels** where the parameters or weights have to be learnt. A filter is a matrix of lower size than the input to it.

The inputs are convolved with the filters and passed through the activation function. The weights of the kernels are randomly initialized and are modified during training based on error-minimization using backpropagation.

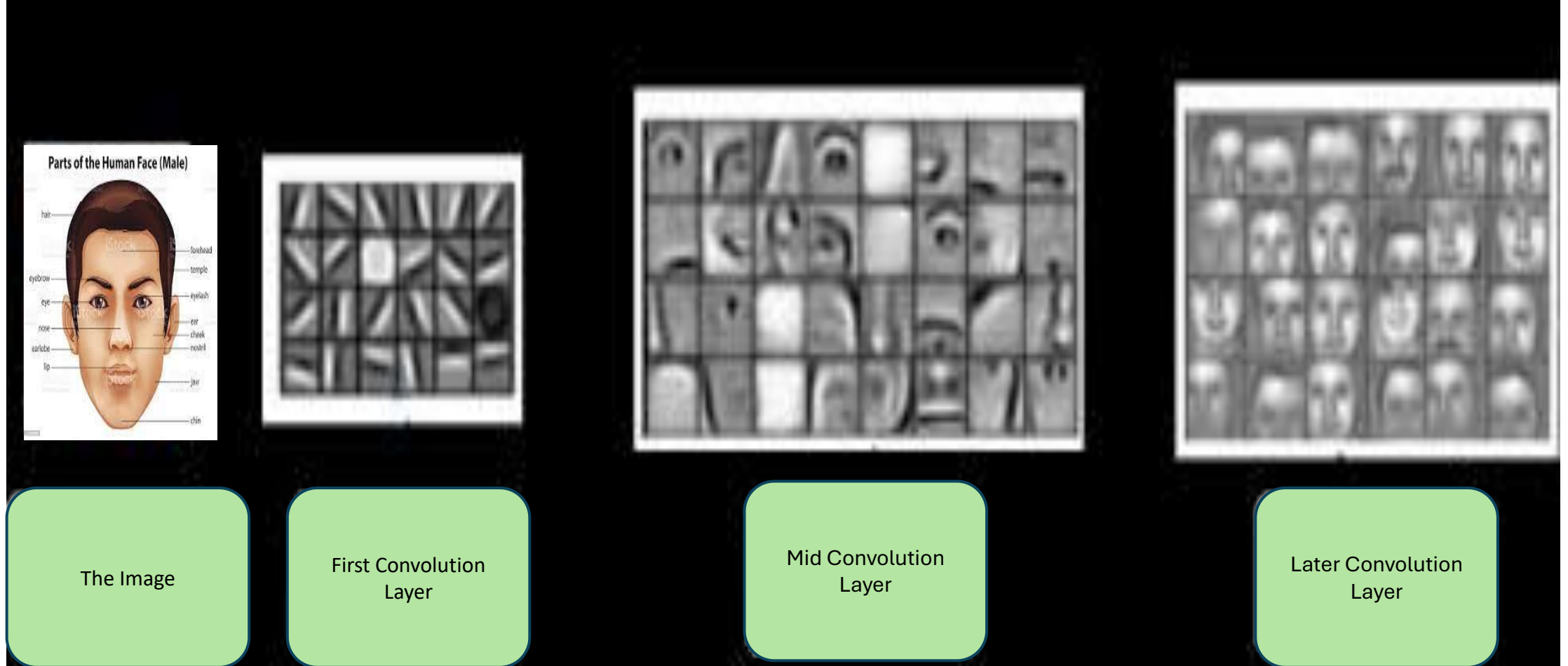
The real values of the **kernel matrix** change with each learning iteration over the training set, indicating that the network is learning to identify which regions are of significance for extracting features from the data.

Stride: The shift of filter after each convolution. It can be increased from 1 to a larger value to decrease overfitting.

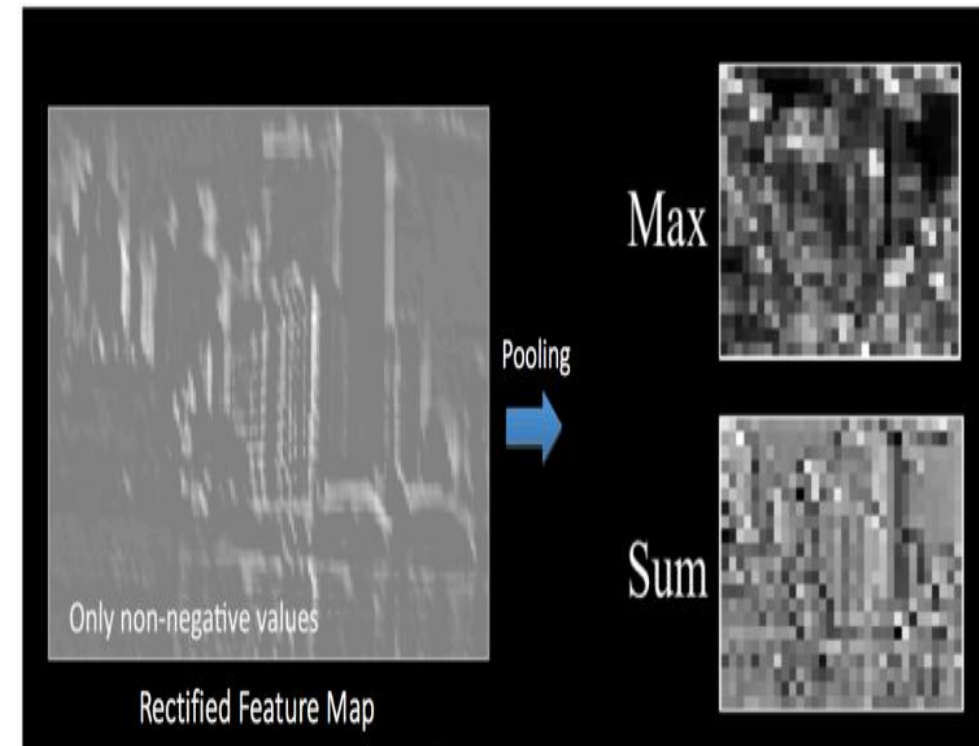
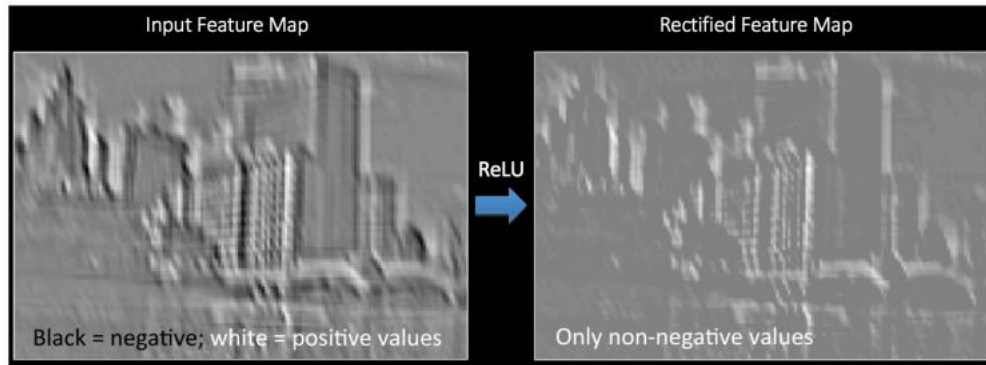
Convolution Neural Network (CNN) - Detailed



Feature Extraction (Intuition In Computer Vision)



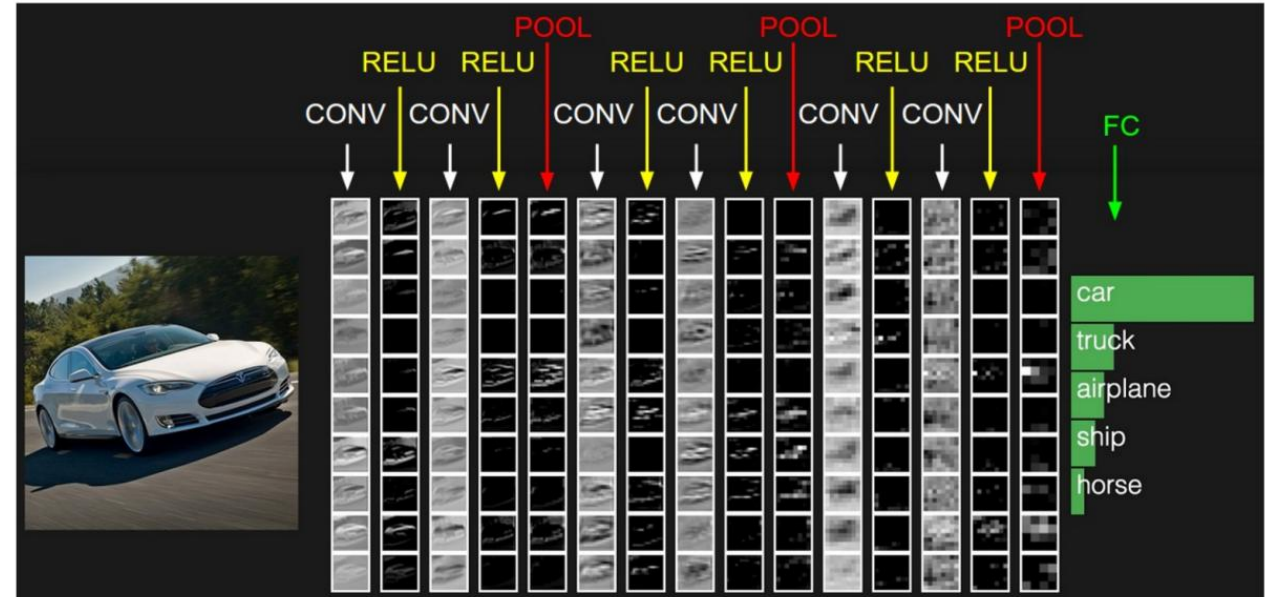
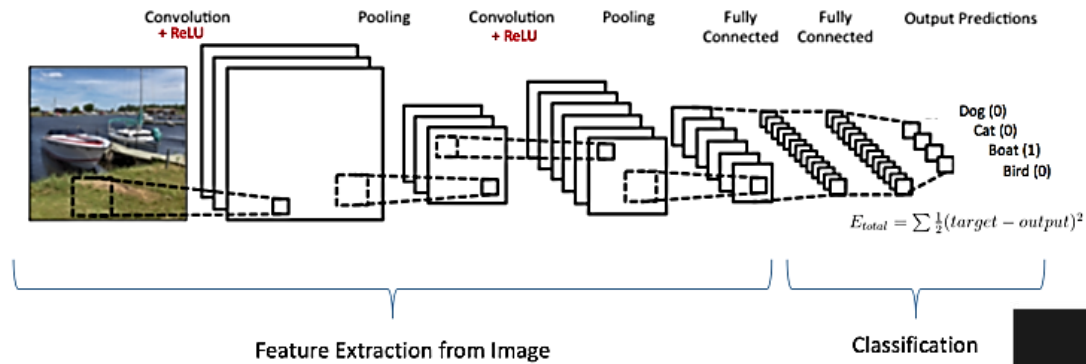
ReLu and Max pooling



Live Demos

- An Interactive Node-Link Visualization of Convolutional Neural Networks
- https://adamharley.com/nn_vis/cnn/2d.html
- **[CNN Playground](#)**
- <https://cnn-playground.live/vision>
- Colab Notebook of Image classification

CNN in summary



Thank you

Happy Learning